Rafay Mahmood

412-520-4745 | mrmahmoo@andrew.cmu.edu | LinkedIn | GitHub | Portfolio

Education

Carnegie Mellon University — Expected May 2027
M.S. in Mechanical Engineering | GPA: 3.84/4.00 — *Pittsburgh, PA*
Thesis: Exoskeleton Control and State Estimation | Coursework: Robot Learning, Optimal Control & RL, Modern Control Theory, Deep Learning

The Pennsylvania State University — May 2025
B.S. in Mechanical Engineering | GPA: 3.49/4.00 — *University Park, PA*

Research and Professional Experience

Graduate Exoskeleton Researcher — September 2025 – Present
MetaMobility Lab - Carnegie Mellon University — *Pittsburgh, PA*

- Built a Python evaluation pipeline comparing model-based and ML joint torque estimators across subjects, speeds (0.4–2.0 m/s), and inclines ($\pm 10°$)
- Built and benchmarked an online model-based human torque estimator (NDO) under real-time data constraints, achieving $R^2 = 0.89$ and 0.127 N·m/kg RMSE against marker-based ground truth with no tuned parameters
- Built a real-time IMU pipeline on an NVIDIA Jetson — five BNO055 sensors over I2C at 44 Hz through a TCA9548A mux and custom integration board — with a VQF-based AHRS and functional sensor-to-segment calibration, achieving 41 ms end-to-end latency
- Diagnosed drift in the BNO055 onboard fusion and moved to raw-sensor mode with an external AHRS, becoming the lab's go-to for BNO055 debugging; traced a torque-estimate artifact to the per-leg chain dynamics
- Derived the Euler-Lagrange dynamics of a 4-DoF foot-rooted biped symbolically and implemented closed-form M/C/N terms in Python, running 178x faster than the numerical model at 74 μs per step

Undergraduate KINARM Robot Researcher — August 2024 – May 2025
Sensorimotor Neuroscience and Learning Lab - The Pennsylvania State University — *University Park, PA*

- Developed a nonlinear Simulink neuromuscular model combining vestibular, proprioceptive, and saccadic eye-movement feedback with an inverted-pendulum controller to simulate responses to mechanical perturbations and generate synthetic data for comparison with KINARM data

eVinci Component Design Intern — May 2024 – August 2024
Westinghouse Electric Corporation — *Cranberry, PA*

- Designed encapsulation segmentation and heat puck models in SolidWorks and Autodesk Inventor Pro, and updated the instrumentation and shielding design document 2 months ahead of internal design review

Projects

Golf Shot Optimization | *Python, MuJoCo, CMA-ES* — Feb 2026 – May 2026

- Built a MuJoCo golf simulation and CMA-ES optimizer that holed shots on two courses within 144 rollouts and cut a hazard course from 3 to 2 strokes via full-horizon sequence planning

Vehicle and Quadrotor Control | *Python, Webots, LQR, MRAC, A** — October 2025 – December 2025

- Designed an integral-augmented LQR lateral controller with PID speed control and A* planning to overtake a slower simulated vehicle, holding 0.50 m mean cross-track error over a full track loop
- Designed an MRAC controller that kept a simulated quadrotor tracking altitude commands through a 67.2% motor thrust loss that crashed the LQR baseline

Controls Education with Maglev | *C++, PID Control, Simulink, SolidWorks* — October 2025 – December 2025

- Led PID controller programming for an IR-sensed maglev system, tuning gains for stable position control from a nonlinear Simulink model of the magnet dynamics, and iterated housing designs in SolidWorks

Technical Skills

Programming: Python, MATLAB, C++
Robotics & ML: Simulink, NumPy, pandas, PyTorch, MuJoCo, OpenSim
Embedded: Linux, NVIDIA Jetson, Teensy 4.1, I2C/CAN
Controls: PID, LQR, MPC, Nonlinear Control, State Estimation (VQF, Madgwick, NDO), Path Planning (A*), Trajectory Optimization, Derivative-Free Optimization (CMA-ES), Adaptive Control (MRAC)
Software: SolidWorks, Autodesk Inventor, KiCad, Git, Vicon Motion Capture